Asset-Liability Risk and the Relationship Between Balance Sheet and Equity Volatility

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Abstract

This paper examines alternative approaches to measuring risk on a financial balance sheet. Using a stylized Asset-Liability Management (ALM) framework, risk is first defined as the volatility of the net return generated by the difference between asset returns and liability costs, measured relative to total assets. The analysis then considers an alternative perspective in which risk is measured relative to equity. It is shown that, although both approaches are based on the same underlying balance-sheet risks, the equity-based measure incorporates a leverage effect that amplifies volatility as equity becomes a smaller share of total assets. Because leverage magnifies volatility, it can also reduce the long-run growth rate of equity relative to what average returns would suggest. A numerical example illustrates how a balance sheet with moderate asset-based risk may exhibit highly volatile equity returns. The results highlight the distinction between evaluating the risk of a balance sheet as a whole and assessing the sensitivity of its capital position to adverse financial shocks.

1 Introduction

From an ALM perspective, the relevant risk measure is often the volatility of the net return relative to total assets. Such a measure may suggest that balance-sheet risk is moderate and well contained. However, this can be misleading when equity represents only a small fraction of the balance sheet. When equity is small relative to total assets, even modest fluctuations in asset and liability values can generate large changes in equity. In this situation, leverage amplifies volatility, causing the long-run growth of equity to fall well below what the average return would suggest. As a result, conclusions drawn from an asset-based risk measure need not carry over to the capital position of the institution. To illustrate this distinction, the analysis below derives the implied volatility of equity returns and compares it with the corresponding balance-sheet risk measure.

2 Risk Measures

Consider a stylized balance sheet in discrete time with no transactions:

$$A_{t+1} = A_t(1 + r_{A,t}), \tag{1}$$
$$L_{t+1} = L_t(1 + r_{L,t}), \tag{2}$$

where A_t denotes assets, L_t interest-bearing liabilities, $r_{A,t}$ the return on assets, and $r_{L,t}$ the funding cost of liabilities. The institution's net position (economic equity) is defined as

$$N_t = A_t - L_t. \tag{3}$$

Define also liabilities as a share of total assets:

$$\alpha_t = \frac{L_t}{A_t}. \tag{4}$$

2.1 Risk measured relative to assets

From an Asset-Liability Management (ALM) perspective, the relevant risk concept is the risk embedded in the consolidated balance sheet. The change in net worth over one period is

$$\Delta N_t = N_{t+1} - N_t = A_t r_{A,t} - L_t r_{L,t}. \tag{5}$$

Dividing by beginning-of-period assets yields

$$\frac{\Delta N_t}{A_t} = r_{A,t} - \alpha_t r_{L,t}. \tag{6}$$

Define the balance-sheet net return as

$$r_{N,t} = r_{A,t} - \alpha_t r_{L,t}. \tag{7}$$

Since both asset returns and liability costs are stochastic, $r_{N,t}$ is also stochastic. Suppressing time subscripts for notational simplicity, the variance of the balance-sheet net return is

$$\mathrm{Var}(r_N) = \sigma_A^2 + \alpha^2\sigma_L^2 - 2\alpha\rho_{AL}\sigma_A\sigma_L, \tag{8}$$

where σ_A and σ_L denote the volatilities of asset returns and liability costs, respectively, and ρ_{AL} their correlation. When liabilities account for nearly all assets, i.e. $\alpha \approx 1$, this expression simplifies to

$$\sigma_N^2 = \sigma_A^2 + \sigma_L^2 - 2\rho_{AL}\sigma_A\sigma_L. \tag{9}$$

If asset and liability returns are perfectly correlated, balance-sheet risk is reduced through natural hedging. In the special case where $\alpha = 1$ and $\sigma_A = \sigma_L$, the balance sheet is perfectly matched and

$$\sigma_N = 0. \tag{10}$$

More generally, the correlation coefficient ρ_{AL} may vary through time. Assets and liabilities often differ in duration, implying that shifts in the yield curve can alter the covariance structure between asset returns and liability costs.

2.2 Risk measured relative to equity

The preceding analysis evaluates risk relative to total assets. An alternative perspective is to measure risk relative to equity itself. The return on equity is defined as

$$r_{E,t} = \frac{N_{t+1} - N_t}{N_t}. \tag{11}$$

Using the definition of net worth,

$$N_t = A_t - L_t, \tag{12}$$

together with

$$N_{t+1} - N_t = A_t r_{A,t} - L_t r_{L,t}, \tag{13}$$

gives

$$r_{E,t} = \frac{A_t r_{A,t} - L_t r_{L,t}}{A_t - L_t}. \tag{14}$$

Substituting $\alpha_t = L_t/A_t$ yields

$$r_{E,t} = \frac{r_{A,t} - \alpha_t r_{L,t}}{1 - \alpha_t} = \frac{r_{N,t}}{1 - \alpha_t}. \tag{15}$$

Thus, the return on equity equals the balance-sheet net return scaled by the inverse of the equity ratio. Suppressing time subscripts, the variance of equity returns becomes

$$\mathrm{Var}(r_E) = \frac{1}{(1-\alpha)^2}\,\mathrm{Var}(r_N). \tag{16}$$

Equivalently,

$$\sigma_E = \frac{\sigma_N}{1-\alpha}. \tag{17}$$

Substituting the expression for σ_N gives

$$\sigma_E = \frac{\sqrt{\sigma_A^2 + \alpha^2\sigma_L^2 - 2\alpha\rho_{AL}\sigma_A\sigma_L}}{1-\alpha}. \tag{18}$$

The underlying balance-sheet risk is identical in both approaches. However, measuring risk relative to equity introduces a leverage factor equal to

$$\frac{1}{1-\alpha}. \tag{19}$$

As liabilities approach total assets and equity becomes small relative to the balance sheet,

$$\alpha \to 1, \tag{20}$$

implying

$$\sigma_E \to \infty, \tag{21}$$

provided that $\sigma_N > 0$. A level of volatility that appears moderate when measured relative to assets may therefore correspond to substantial volatility when measured relative to equity.

2.3 Numerical illustration

To illustrate the difference between balance-sheet risk and equity risk, consider the following stylized example. Let

$$A = 100, \tag{22}$$
$$L = 90, \tag{23}$$

implying that

$$\alpha = \frac{L}{A} = 0.9 \tag{24}$$

and

$$N = A - L = 10. \tag{25}$$

Assume further that the volatility of asset returns is $\sigma_A = 8\%$, the volatility of liability costs is $\sigma_L = 2\%$ and that the correlation between asset returns and liability costs is $\rho_{AL} = -0.9$. The volatility of the balance-sheet net return is therefore

$$\begin{aligned}
\sigma_N^2 &= \sigma_A^2 + \alpha^2\sigma_L^2 - 2\alpha\rho_{AL}\sigma_A\sigma_L && (26)\\
&= 0.08^2 + 0.9^2 \times 0.02^2 - 2 \times 0.9 \times (-0.9) \times 0.08 \times 0.02 && (27)\\
&= 0.009316. && (28)
\end{aligned}$$

Hence,

$$\sigma_N = \sqrt{0.009316} = 0.0965. \tag{29}$$

The volatility of the net return measured relative to assets is thus approximately

$$\boxed{\sigma_N \approx 9.65\%.} \tag{30}$$

Since the equity ratio is

$$1 - \alpha = 0.1, \tag{31}$$

the volatility of equity growth becomes

$$\sigma_E = \frac{\sigma_N}{1-\alpha} \tag{32}$$

$$= \frac{0.0965}{0.1} \tag{33}$$

$$= 0.965. \tag{34}$$

Thus,

$$\boxed{\sigma_E \approx 96.5\%.} \tag{35}$$

The same result can be interpreted in currency units. Since total assets equal 100, the standard deviation of the annual change in the net position is

$$100 \times 0.0965 = 9.65. \tag{36}$$

This amount is almost equal to the entire equity position of 10. Consequently, even though the volatility of the balance-sheet net return is below 10%, the volatility of equity returns is close to 100%. The example illustrates the distinction between the two risk concepts. Measured relative to assets, the balance-sheet risk appears moderate. Measured relative to equity, however, the same underlying balance-sheet risk becomes highly amplified because equity constitutes only a small fraction of the balance sheet. The amplification factor is given by

$$\frac{1}{1-\alpha} = 10. \tag{37}$$

This leverage effect explains why institutions with relatively small capital buffers may exhibit very high volatility in equity returns, even when the underlying balance-sheet risk is modest.

2.4 Volatility drag and the distribution of equity outcomes

The previous analysis focused on volatility as a measure of risk. However, *when leverage is high, volatility affects not only dispersion but also the long-run growth rate of equity.* Consider again the stylized balance sheet introduced above. Even if the expected return on assets exceeds the funding cost of liabilities, fluctuations in asset and liability values may reduce the long-run growth rate of equity. This phenomenon is commonly known as *volatility drag.* For a multiplicative process with return volatility σ, the long-run geometric growth rate is approximately

$$g \approx \mu - \frac{1}{2}\sigma^2, \tag{38}$$

where μ denotes the arithmetic mean return. Since

$$\sigma_E = \frac{\sigma_N}{1-\alpha}, \tag{39}$$

the volatility drag on equity increases rapidly as leverage rises. Consequently, the most likely long-run equity outcome may be substantially below the arithmetic mean outcome, even when expected returns remain positive.

To illustrate the interaction between leverage and volatility drag, a Monte Carlo simulation was performed for a stylized balance sheet with assets equal to 100, liabilities equal to 90, and equity equal to 10. Asset returns were assumed to have mean 2.25% and volatility 8%, while liability costs had mean 2% and volatility 2%. The correlation between asset and liability returns was set to -0.9. The balance sheet was simulated over a five-year horizon using 500,000 scenarios. Figure 1 reports the resulting distribution of return on equity (ROE) outcomes after five years.

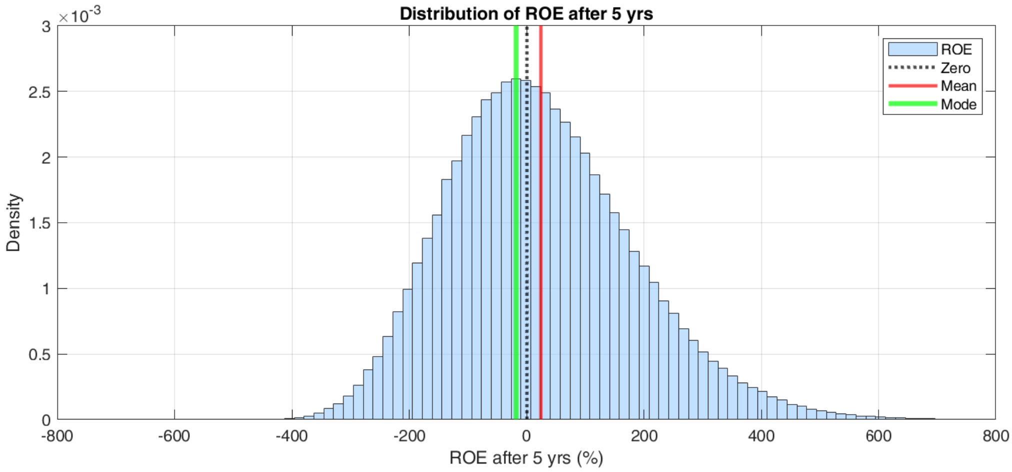

Figure 1: Simulation of ROE

The resulting asymmetry of equity outcomes is illustrated in Figure 1. As leverage increases, a small number of highly favourable outcomes raise the arithmetic mean, while the majority of realizations remain concentrated at lower levels of equity. As a consequence, the most likely outcome may differ substantially from the expected outcome and may even become negative despite positive expected returns. The distinction between the mean, median and mode is particularly important in the presence of volatility drag. While the arithmetic mean is boosted by a small number of very favourable outcomes, the median and especially the mode are more representative of the outcome experienced in a typical simulation. Under high leverage, it is therefore possible for the expected return on equity to remain positive even though the most likely outcome, as measured by the mode, is close to or below zero.

3 Conclusion

The balance-sheet approach and the equity approach address fundamentally different aspects of financial risk. The former measures the risk embedded in the institution's overall asset-liability structure, whereas the latter measures how strongly that risk affects the capital base available to absorb losses. As demonstrated above, the two perspectives can lead to markedly different conclusions. A level of volatility that appears moderate when measured relative to total assets may correspond to substantial volatility when measured relative to equity. This difference arises from the leverage effect inherent in the balance sheet and becomes particularly pronounced when equity represents only a small share of total assets. The appropriate risk measure therefore depends on the question being asked. A balance-sheet perspective is well suited for assessing the economic risk of the institution as a whole, while an equity-based perspective is more informative when evaluating capital resilience and the ability to absorb adverse shocks. Taken together, the two approaches provide complementary views of financial risk and should be interpreted jointly rather than as substitutes.